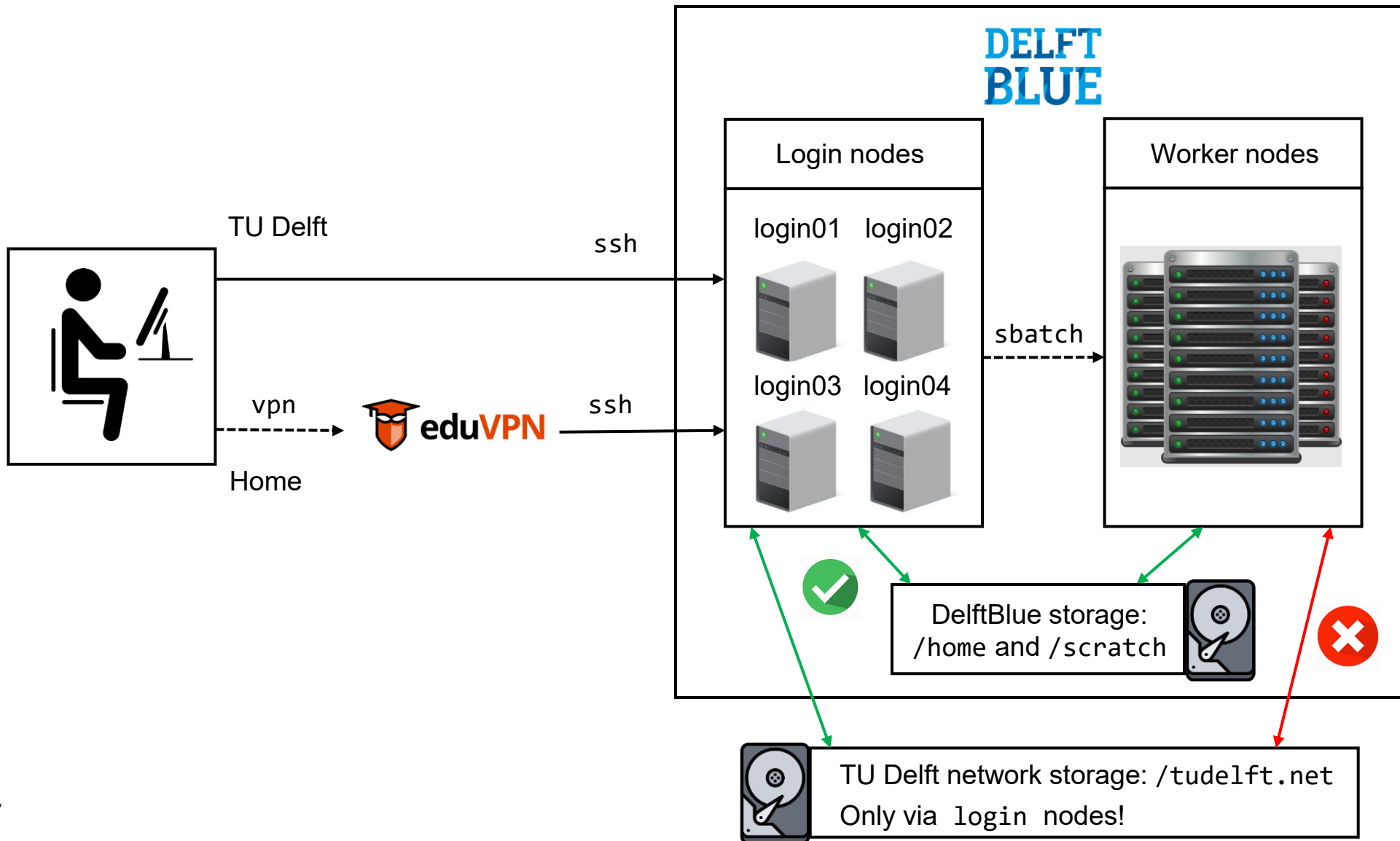


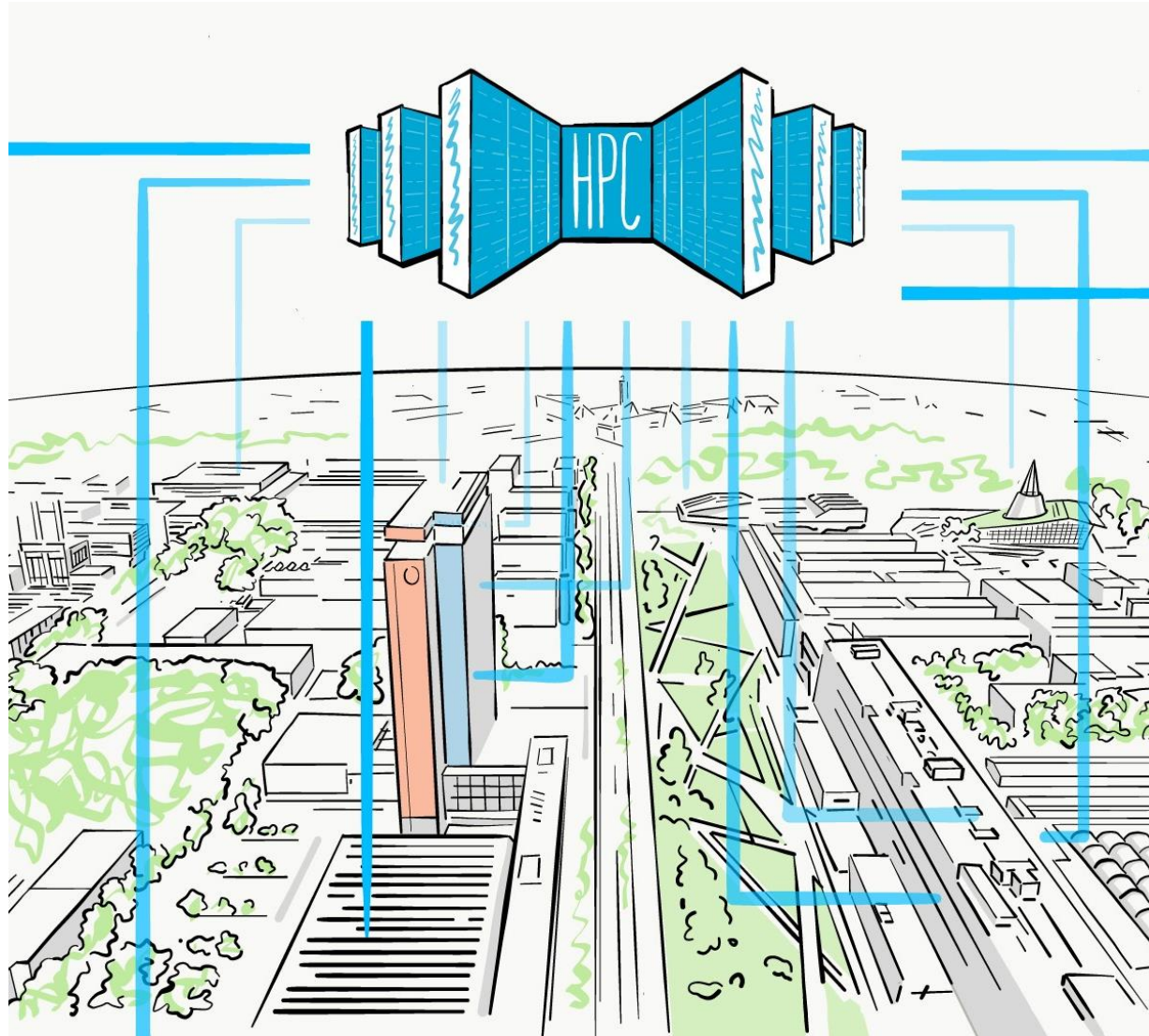


DelftBlue

- 16.000 CPU cores
- Over 300 compute nodes
- 20 GPU nodes



DelftBlue community: 3mE, ABE, AE, AS, CEG, EEMCS, IDE, TPM, QuTech



How do I work with cluster?

1. **Prepare input files for your code** on a personal computer
2. **Upload input files** and required data to the cluster's storage
3. **Determine** required **resources**
4. **Create job script**
5. **Submit** job(s) to scheduler
6. **Monitor** progress (via output files) and resource use (via statistics)
7. **Download** results to personal computer for further processing
8. **Cleanup** files

Typical submission script

- Create a job script in a file

```
#!/bin/sh

#SBATCH --job-name=job_name
#SBATCH --partition=compute
#SBATCH --account=research-eemcs-diam
#SBATCH --time=01:00:00
#SBATCH --ntasks=4
#SBATCH --cpus-per-task=1
#SBATCH --mem-per-cpu=1GB

module load 2026
module load cpu
module load openmpi
srun ./executable > output.log
```

- system info: type of script
- select job name
- select partition to run your job on
- specify account
- request run time
- number of tasks (parallel)
- CPUs (threads) per task
- RAM per CPU
- Use DelftBlue software collection
- Enable CPU-compiled stack
- load openmpi module
- start tasks with srun

Typical scheduler commands?

- Submit job using the job script

```
$ sbatch jobscript.sbatch
Submitted batch job 290573
```

- See queue status

```
$ squeue
JOBID PARTITION      NAME      USER ST  TIME
290573  general    jobscrip  somebody  R   0:01
```

- See job output

```
$ cat slurm-290573.out
Hello world!
```

- Cancel job

```
$ scancel 290573
$ squeue
JOBID PARTITION      NAME      USER ST  TIME
NODES NODELIST(REASON)
```

What is a scheduler?

```
2753413 compute KUS9e0 emadaï R 1-10:45:10 12 cmp[025,028,053,062,104-106,157,182,191,196-197]
2753412 compute KUS8e0 emadaï R 1-11:26:21 5 cmp[062,121,123,181,184]
2753410 compute KUS7e0 emadaï R 1-11:32:23 4 cmp[036,055-056,216]
2755343 compute gaT4War nikosvas R 16:10 1 cmp217
2753385 compute US11e000 emadaï R 1-14:34:15 13 cmp[010,016,021,030,037,054,069-070,158-159,195,212-213]
2753356 compute US10e000 emadaï R 1-18:45:28 21 cmp[002,018,065,084,087,095,101-103,105-106,109-110,125-127,129,162-164,182]
2753362 compute US11e000 emadaï R 1-18:43:25 9 cmp[016,109-110,134-136,174-176]
2753353 compute US9e000 emadaï R 1-19:31:59 12 cmp[008,019,066,077,093,115,118,134,150,163,181,183]
2753349 compute US6e000 emadaï R 1-19:42:11 6 cmp[097,173-175,190-191]
2753340 compute US0e000 emadaï R 1-19:45:14 10 cmp[046,107-110,124,173,201-203]
2753350 compute US7e000 emadaï R 1-19:42:11 13 cmp[009,025-026,030,082,085,093,099,109-110,129,161,195]
2753352 compute US8e000 emadaï R 1-19:40:09 8 cmp[009,011-012,064-065,207-209]
2753342 compute US1e000 emadaï R 1-19:44:13 5 cmp[169-173]
2753343 compute US2e000 emadaï R 1-19:44:13 11 cmp[036-039,157-159,204-207]
2753345 compute US3e000 emadaï R 1-19:44:13 5 cmp[085,140,199,215-216]
2753346 compute US4e000 emadaï R 1-19:43:12 7 cmp[061-064,115,136-137]
2753348 compute US5e000 emadaï R 1-19:43:12 5 cmp[003,014,058,088,093]
2753337 compute USm1e000 emadaï R 1-22:27:34 10 cmp[041-044,084,113,122-123,178,210]
2755699 compute ms-5_T_2 darshanr R 27:24 1 cmp122
2755698 compute ms-5_T_2 darshanr R 52:57 1 cmp158
2755697 compute ms-5_T_2 darshanr R 2:31:26 1 cmp012
2755696 compute ms-5_T_2 darshanr R 2:32:27 1 cmp201
2755695 compute ms-5_T_2 darshanr R 3:23:31 1 cmp122
2755694 compute ms-5_T_2 darshanr R 3:45:08 1 cmp022
2755692 compute ms-5_T_2 darshanr R 3:57:28 1 cmp144
2755693 compute ms-5_T_2 darshanr R 3:55:24 1 cmp025
2749288 compute M11 ydai2 R 4-01:45:13 50 cmp[004-006,016-019,026,047-055,066-067,069-070,074-076,084,104,107,109-110,114-115,
131-132,160-162,168-169,171-172,182,190,194-195,202-204,213,217-218]
2749330 compute M110 ydai2 R 4-01:35:12 36 cmp[007-011,020-022,072-073,081-082,091,095-099,116-120,152-157,163-166,210-212]
2757611 compute fmT4man nikosvas R 37:36 1 cmp167
2757612 compute fmT4war nikosvas R 36:35 1 cmp195
2753175 compute IPMC atajeddi R 2-00:47:23 4 cmp[116-119]
2756707 compute Q1 jjgreep R 9:55:20 1 cmp198
2756838 compute mcts_dis gmeppeli R 9:45:42 1 cmp041
2756448 compute random-w sajvanle R 11:20:27 1 cmp044
2756447 compute random-5 sajvanle R 12:11:13 1 cmp069
2756446 compute random-w sajvanle R 12:21:51 1 cmp134
2756445 compute random-w sajvanle R 12:28:16 1 cmp187
2756444 compute random-4 sajvanle R 13:45:02 1 cmp180
2757149 compute t_bin arodrigu R 15:05:59 1 cmp177
2757291 compute param arodrigu R 10:08:51 1 cmp068
2756837 compute mcts_dis gmeppeli R 18:15:47 1 cmp160
2756836 compute mcts_dis gmeppeli R 18:36:56 1 cmp131
2756835 compute mcts_dis gmeppeli R 18:38:00 1 cmp131
2756834 compute mcts_dis gmeppeli R 18:43:18 1 cmp159
```

Module system demo

```
[NetID@login02 ~]$ module avail
```

→ list available modules

```
[NetID@login02 ~]$ module load
```

→ load module

```
[NetID@login02 ~]$ module list
```

→ list loaded modules

```
[NetID@login02 ~]$ module spider {module}
```

→ find module {module}

More info: <https://doc.dhpc.tudelft.nl/delftblue/DHPC-modules/>

Exercises on GitLab:

```
git clone https://gitlab.tudelft.nl/diam/delftblue-intro
```

References

- Documentation:

<https://www.tudelft.nl/dhpc/documentation>

- Mattermost:

<https://mattermost.tudelft.nl/dhpc/>

- Self Service Portal (TopDesk):

<https://tudelft.topdesk.net>

- DHPC Courses:

<https://www.tudelft.nl/dhpc/courses>

Courses

Command Line and
DelftBlue Basics

High-performance
computing using
OpenMP

Introduction into
Programming using
MPI

Introduction to Julia

Machine learning
using PyTorch on
DelftBlue

Parallel computing
with Python on
DelftBlue

Programming on the
GPU with CUDA

Planned courses

25 SEPTEMBER 2024 09:30 T/M
16:30

Command line and DelftBlue
basics 25 September 2024



15 OKTOBER 2024 09:30 T/M 18
OKTOBER 2024 16:30

Introduction to Julia for High-
Performance Computing



18 OKTOBER 2024 09:30 T/M
17:00

Introduction into Programming
using MPI



25 OKTOBER 2024 09:30 T/M
16:30

Command line and DelftBlue
basics 25 October 2024



08 NOVEMBER 2024 08:30 T/M
18:00

DHPC User Summit



11 NOVEMBER 2024 09:30 T/M 12
NOVEMBER 2024 16:30

Programming on the GPU with
CUDA



04 DECEMBER 2024 09:30 T/M
16:30

Machine learning using PyTorch
on DelftBlue 10 April 2024



05 DECEMBER 2024 09:30 T/M
16:30

Command line and DelftBlue
basics 5 December 2024

